Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/02/2024
PI Name:	Diaz Artiles, Ana Ph.D.		
Project Title:	Effects of Altered-Gravity on Perception and Bi-manual Coordination: Impacts on Functional Performance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	08/01/2020	**End Date:**	07/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	Parabolic		
Flight Assignment:	NOTE: End date changed to 7/31/2025 per NSSC information (Ed., 9/22/24) NOTE: End date changed to 7/31/2024 per NSSC information (Ed., 6/15/22) NOTE: End date changed to 7/31/2022 per NSSC information (Ed., 7/6/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dunbar, Bonnie Ph.D. (Texas A&M University) Kennedy, Deanna Ph.D. (Texas A&M University)		
Grant/Contract No.:	80NSSC20K1499		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Many of the activities associated with spaceflight require individuals to use both limbs simultaneously to accomplish the task. Motor control, as well as visual performance and spatial orientation are disrupted by gravitational transitions between 1 G and 0 G, but very little is known about the sensorimotor deficits between 0 G and 1 G. The objective of this analog-based research effort is to investigate the impact of partial G-levels on bimanual coordination tasks that are operationally relevant for spaceflight. The same set of human subjects will participate in two different bimanual coordination tasks during parabolic flight, which will deliver G-levels of 0, 0.25, 0.5, 0.75, 1, and 1.8 G. Sensorimotor dose-response curves will be generated between bimanual coordination operational variables as a function of G-level, and G-thresholds (which indicate when performance decrements occur) will be determined. We will also quantify the risk associated with the use of a common motion sickness drug (promethazine) during bimanual coordination tasks. Results will provide critical information for current and future countermeasure development and in-flight prescriptions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project investigates the influence of gravity on bimanual coordination using a variety of altered-gravity analogs. Results will provide critical information for current and future sensorimotor-related countermeasures and in-flight prescription. In addition, this research effort has direct application to bimanual coordination tasks on Earth, for example during complex tasks that require a coordinated two-limb movement, as well as for rehabilitation purposes.
Task Progress:	At the end of year 4, we completed data collection during the parabolic flight campaign in June 2023 and began preliminary data analysis. About 1 week before the flights, we sent a forward team to Bordeaux with all the hardware required for data collection. The forward team assembled and debugged our data acquisition system to ensure there had been no damage to equipment during travel. Hardware was then set up in accordance with Novespace's safety regulations – which included securing equipment to their designated baseplates. Once the setup passed their safety inspection, we moved the hardware onto the plane to finish the setup and make any final adjustments necessary to facilitate smooth data collection. Some small changes were made to the data acquisition software in LabVIEW to minimize the potential for operator error. Next, we finalized experiment procedures (including baseline, pre/post flight, and in-flight) in preparation for flight rehearsals upon the rest of the team's arrival. The team drilled on the experiment protocol with all operators prior to any data collection. We successfully trained participants and collected baseline one day before any flights. In addition, we successfully collected pre-flight, in-flight, and post-flight data over the next 4 days. We are currently conducting data analysis to address our aims and research questions, and we include our preliminary results in the present report. We are analyzing measures of coordination, motion sickness, and EMG-EMG coherence to identify significant factors altering performance and generating dose-response curves. In addition, we continue to investigate the bimanual coordination disruption (and adaptation) following gravitational transitions experienced during the parabolic flight. The NASA Institutional Review Board (IRB) for parabolic flight has been updated to reflect that the study is closed for enrollment, we have completed the collection of data and information from 12 participants, completed the parabolic flights, and that current study activities are limited to data analysis. We are in the process of updating the TAMU IRB as well to show these changes. This will give us the flexibility to reopen our IRBs to new enrollment with minimal paperwork overhead if we have the opportunity to test in another parabolic flight campaign in the future. During year 4, we have also continued our ground experiments efforts. We completed an additional ground study using Lower Body Negative Pressure (LBNP), and we continue to analyze our results from this, and the other ground studies already conducted. We published an additional journal paper and several other abstracts, and other publications are in progress. In addition, during next year, we will continue to investigate using centrifugation.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Diaz-Artiles A, Keller N, Weinrich M, Abbott R, Wright T, Dunbar B, Kennedy D. "Bimanual coordination during partial gravity in parabolic flight: Preliminary results." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstract. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Kennedy D, Neto O, Weinrich M, Keller N, Abbott R, Wright T, Dunbar B, Diaz-Artiles, A. "EMG-EMG wavelet coherence of muscle coupling during bimanual tasks performed in parabolic flight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. EMG-EMG wavelet coherence of muscle coupling during bimanual tasks performed in parabolic flight. , Feb-2024
Abstracts for Journals and Proceedings	Kennedy DM, Neto OP, Weinrich MM, Wang Y, Abbott R, Keller N, Wright T, Dunbar B, Diaz-Artiles, A. "The effects of inherent and incidental constraints on bimanual force control in parabolic flight." 2024 Society for the Neural Control of Movement, Dubrovnik, Croatia, April 14-19, 2024. Abstract. 2024 Society for the Neural Control of Movement, Dubrovnik, Croatia, April 14-19, 2024. , Apr-2024
Articles in Peer-reviewed Journals	Strohm AO, Johnston C, Hernady E, Marples B, O'Banion MK, Majewska AK. "Cranial irradiation disrupts homeostatic microglial dynamic behavior." J Neuroinflammation. 2024 Apr 3;21:82. https://doi.org/10.1186/s12974-024-03073-z ; PubMed PMID: 38570852; PubMed Central PMCID: PMC10993621 , Apr-2024
Articles in Peer-reviewed Journals	Wang Y, Neto OP, Weinrich M, Abbott R, Diaz-Artiles A, Kennedy DM. "The effect of inherent and incidental constraints on bimanual force control in simulated Martian gravity." Hum Mov Sci. 2024 Jun;95:103199. https://doi.org/10.1016/j.humov.2024.103199 , Jun-2024